

Anish Guntreddi

479-402-4854

<https://personal-website-wine-one.vercel.app>

<https://github.com/Anish-Guntreddi>

<https://www.linkedin.com/in/anish-guntreddi/>

EDUCATION

University of Miami

Coral Gables, Florida

B.S: Data Science & AI, Computer Science, Mathematics

Expected: May 2026

Honors: Stamps Scholar at University of Miami (Highest Honor Scholarship at University of Miami), Foote Fellow Honors

SKILLS

Programming Languages: Python, HTML, CSS, Java, JavaScript, C++, C#, C, Swift, SQL, PHP, Excel VBA

Frameworks: NextJs, ReactJs, TensorFlow, Pytorch, NodeJS, TailwindCSS, Laravel, MySQL, Mongoose, Express, Xcode, React Native

Services / Databases: Clerk, MongoDB, Supabase, Firebase, MariaDB, Swagger, Docker

WORK EXPERIENCE

Walmart Global Technology

Coral Gables, Florida

Machine Learning Engineer and Software Engineer Intern

May 2025 – Till Date

- Developed AI agents using LangChain, LangGraph, and Next.js, delivering full end-to-end deployment capabilities with modular backend and frontend integration..
- Designed and implemented a RAG pipeline leveraging the Milvus vector database for embedding storage and retrieval, enhancing query accuracy across applications.

University of Miami Orthopedics Research Labratory

Coral Gables, Florida

Student Machine Learning Engineer and Researcher

March 2025 – Till Date

- Designed and trained a Convolutional Neural Network (CNN) on Electromyography (EMG) signals to support muscle rehabilitation and recovery monitoring.
- Implemented signal preprocessing (filtering, normalization) and data augmentation techniques to enhance model robustness and generalization.
- Collaborated with clinical researchers to align model outputs with rehabilitation protocols, enabling data-driven insight for patient progress tracking.

Insigneo Financial Group

Coral Gables Florida

Machine Learning Engineer and Quantitative Analyst

October 2024 – December 2024

- Engineered and preprocessed large-scale financial time-series datasets, incorporating factor modeling and technical indicators for AI-driven portfolio optimization.
- Applied machine learning techniques (feature selection, model tuning) and conducted market research to evaluate strategies optimizing performance based on Sharpe ratio, drawdown, and risk-adjusted returns.

DUCAB (Dubai Cable Company)

Dubai, United Arab Emirates

Quantitative Analyst and Data Science Intern

June 2024 – August 2024

- Designed and optimized a proprietary algorithmic trading bot to enhance portfolio performance through backtesting and parameter tuning.
- Built a Python-SAP integration pipeline to extract purchase order data from PDFs and automate entry, reducing processing time by 80%.

University of Miami Cancer Research Laboratory

Coral Gables, Florida

Student Researcher | R | Rstudio | Python |

November 2023 – April 2024

- Conducted bioinformatics research on the correlation between HERV gene expression and various cancers, with a focus on glioblastoma multiforme.
- Analyzed multi-region patient gene expression datasets in R and Python, generating volcano plots and other statistical visualizations to identify significant differential expression patterns.

AVAS Technology

Atlanta, Georgia

Full Stack Software Engineering Intern | Laravel | PHP | JUnit | MariaDB |

August 2023 – December 2023

- Designed and implemented RESTful APIs, integrated JUnit testing for continuous quality assurance, and led backend migration to support an enhanced classroom management system.
- Created and maintained a full stack web platform with PHP, Laravel, and MariaDB with modular backend architecture

PROJECTS

Unsupervised Machine Learning Algorithmic Trading Bot

July 2024

| Python | Jupyter Notebook |

- Developed a K-Means clustering-based trading model to identify high-growth S&P 500 equities, leveraging technical indicators (ATR, MACD, Bollinger Bands, RSI), Fama-French factors, and Garman-Klass volatility estimates as predictive features.
- Classified equities into growth, value, and neutral clusters through quantitative market research, leveraging factor exposures and volatility profiles to inform portfolio construction.

Full Stack Real Estate Project

July 2024

- Built a full-stack real estate platform with Next.js, Clerk authentication, and Supabase, implementing secure CRUD operations for property management.
- Integrated Google Maps API for dynamic geospatial search and location-based filtering, with a responsive HTML/CSS UI for optimized user experience.

Text-Report AI Agent

Summer 2025

- Embedded Walmart-specific databases into a vector database to enable context-aware querying for a text-based business intelligence assistant.
- Designed and implemented a RAG pipeline using LangChain and LangGraph to answer business questions across multiple operational domains
- Developed both the agentic AI workflow and the Next.js frontend, integrating backend inference with an intuitive user interface.